
POWER BI CASE STUDY

Incident Performance & Operational Analysis

SLA PERFORMANCE | RESOLUTION EFFICIENCY | WORKLOAD | OPERATIONAL RISK

IT SERVICE MANAGEMENT

INCIDENT ANALYSIS

Edward Jon Arquiza

Data Analyst Portfolio

2026

1. Executive Summary

Incident Performance & Operational Analysis

This case study evaluates IT service management incident performance using Power BI, with a focus on **incident workload, SLA compliance, resolution efficiency, reassignment activity, priority, category, subcategory, and assignment-group performance.**

The analysis was designed to help an IT service organization understand where operational demand is concentrated, where service-level performance is under pressure, and which areas may require further investigation or improvement.

The analysis identified **24,918 incidents** within the analyzed dataset. Overall SLA success was **63.42%**, meaning **36.58% of incidents did not meet SLA**. Mean time to resolve was **178.17 hours**, while median time to resolve was considerably lower at **22.10 hours**. This difference indicates that a smaller number of long-running incidents have a substantial effect on the average resolution time.

Reassignment was also a significant operational signal. Approximately **45.63% of incidents were reassigned**, indicating that a substantial portion of the workload passed through more than one assignment before resolution.

The dashboard further shows that operational performance varies considerably across **incident priorities, categories, subcategories, and assignment groups**. High incident volume identifies where workload is concentrated, while SLA performance, resolution time, and reassignment rate provide additional context for determining where operational attention may be required.

Key Findings

- **24.918K total incidents** — the organization processed a substantial volume of incident demand during the analyzed period.
- **63.42% SLA success rate** — approximately one-third of incidents did not meet the defined SLA, making SLA performance an important operational improvement area.
- **36.58% missed SLA rate** — the size of the missed-SLA population warrants further investigation into the operational factors associated with SLA breaches.
- **178.17-hour mean resolution time vs. 22.10-hour median** — the large difference suggests that a smaller number of very long-running incidents materially influence the overall average.
- **45.63% reassignment rate** — nearly half of incidents were reassigned, making incident routing and ownership an important area for operational review.
- **Priority performance varies substantially** — lower-priority incidents show considerably stronger SLA performance than high- and critical-priority incidents in the analyzed data.

- **Workload is concentrated** — a relatively small number of categories and subcategories account for a large portion of total incident volume, creating opportunities to target recurring sources of demand.
- **Assignment-group performance varies** — reassignment and SLA performance differ significantly between groups, allowing management to identify areas requiring closer operational investigation.

Primary Recommendation

The primary recommendation is to **focus operational improvement efforts on SLA performance, long-running incidents, and reassignment activity**, while using incident volume and priority to determine where the greatest potential impact exists.

Rather than optimizing a single KPI, management should evaluate **SLA performance + resolution time + reassignment rate + incident volume + priority** together.

This provides a more balanced view of operational performance and helps distinguish between areas that simply handle large workloads and areas where the workload is also associated with service-performance risk.

The goal is not simply to identify where incidents occur, but to identify where operational effort can produce the greatest improvement.

2. Business Scenario & Objectives

2.1 Business Scenario

For this portfolio project, I assumed the role of a **Data Analyst supporting an IT Service Management organization**.

The organization manages a large volume of IT incidents across different categories, priorities, assignment groups, and service areas. While incident-level data is available, management needs a clearer view of **service performance, SLA compliance, resolution efficiency, workload concentration, and operational friction**.

The analysis uses incident records containing information such as incident priority, category, subcategory, assignment group, SLA status, reassignment activity, reopen activity, and resolution time.

The project is presented as a **simulated business case for portfolio purposes**.

Management wants to understand:

- How large is the overall incident workload?
- How effectively are incidents meeting SLA?

- How quickly are incidents being resolved?
- Which priorities have the greatest SLA pressure?
- Which categories and subcategories generate the most incident volume?
- Which assignment groups handle the greatest workload?
- Where are reassignment rates particularly high?
- Which assignment groups combine poor SLA performance with long resolution times?
- Where is operational workload concentrated across categories and priorities?
- Which areas should management prioritize for further investigation and improvement?

2.2 Business Objective

The primary objective was to transform incident-level IT service management data into an **interactive Power BI dashboard that helps management understand service performance, identify operational pressure points, and support data-driven improvement decisions.**

The analysis focuses on three major areas:

1. Service Performance

Evaluate:

- SLA success rate
- Missed SLA rate
- Mean resolution time
- Median resolution time
- Resolution-time distribution
- SLA performance by priority

2. Operational Workload

Evaluate:

- Total incident volume
- Incident volume by category
- Incident volume by subcategory
- Incident volume by assignment group
- Priority and category workload concentration

3. Operational Friction & Risk

Evaluate:

- Reassignment rate

- Reassignment performance by assignment group
- SLA performance versus resolution time
- Assignment groups requiring attention
- Areas where workload and service-performance risk overlap

2.3 Key Business Questions

The analysis was designed to answer the following questions:

1. **How large is the organization's incident workload?**
2. **What percentage of incidents meet the defined SLA?**
3. **How significant is the missed-SLA population?**
4. **How quickly are incidents being resolved, and how different are mean and median resolution times?**
5. **Which priority levels experience the greatest SLA pressure?**
6. **Which incident categories generate the most workload?**
7. **Which subcategories contribute the greatest incident volume?**
8. **Which assignment groups experience the highest reassignment rates?**
9. **Which assignment groups combine low SLA performance with slower resolution times?**
10. **Where is incident workload concentrated across category and priority?**
11. **Which areas should management investigate first to improve service performance and operational efficiency?**

These questions were intentionally designed to move beyond simple incident counting. The goal is to connect **volume, service performance, resolution efficiency, and operational behavior** so that the dashboard can support prioritization rather than simply describe the dataset.

2.4 Scope

The analysis covers the incident records available within the selected dashboard analysis period.

The Power BI report provides a shared **Date Range** filter and **Assignment Group** filter, allowing users to interactively explore the incident population.

The analysis considers:

Incident workload

- Incident volume
- Category
- Subcategory
- Assignment group

Service performance

- SLA success
- SLA misses
- Resolution time
- Priority

Operational behavior

- Reassignments
- Reopened incidents
- Assignment-group performance

The analysis is intended to identify **patterns and areas requiring further investigation**. Where the available data shows an association or operational signal, it is not automatically treated as proof of an underlying cause.

2.5 Stakeholder Perspective

The primary audience for this analysis is assumed to be **IT service management leadership and operational managers**.

Different stakeholders can use the dashboard for different purposes:

Stakeholder	Potential Use
IT Service Management Leadership	Monitor overall SLA and incident performance
Service Desk Managers	Identify workload and reassignment pressure
Assignment Group Managers	Compare group-level performance
Operations Managers	Investigate resolution bottlenecks
Process Owners	Identify recurring categories and subcategories
Analysts	Investigate operational patterns and support improvement initiatives

The dashboard therefore prioritizes **clear KPIs, benchmarks, workload concentration, and actionable comparisons** rather than purely technical reporting.

3. Data & Preparation

3.1 Dataset

The analysis uses an **IT Service Management incident dataset** containing incident-level records from an IT support environment.

For this portfolio project, the dataset was treated as a **simulated business data source**. The objective was to transform the raw incident records into a structured analytical model capable of supporting analysis of **incident workload, SLA performance, resolution efficiency, reassignment activity, priority, category, subcategory, and assignment-group performance**.

The primary incident table contains operational, classification, SLA, and resolution information, including fields such as:

- Incident number
- Incident state
- Active status
- Reassignment count
- Reopen count
- System modification count
- SLA status
- Caller
- Opened date/time
- Contact type
- Location
- Category
- Subcategory
- Impact
- Urgency
- Priority
- Assignment group
- Assigned-to
- Knowledge usage
- Reassignment-related fields
- Problem and change references
- Resolution information
- Closed information
- Resolution time
- Closure time
- Resolution-to-closure time

The raw structure therefore provided sufficient information to analyze both **service performance** and **operational workload**.

3.2 Data Structure

The main transactional table used for the analysis was:

Incidents

The Incidents table was retained as the primary incident-level fact table.

It contains one record per incident and provides the foundation for:

- Incident volume
- SLA analysis
- Resolution-time analysis
- Reassignment analysis
- Reopen analysis
- Priority analysis
- Category analysis
- Subcategory analysis
- Assignment-group analysis

The incident table was kept at its original transactional grain rather than being unnecessarily flattened or duplicated.

Supporting dimension tables were later created for important descriptive attributes, including:

- **Dim_Category**
- **Dim_Subcategory**
- **Dim_Assignment_Group**

These dimensions were used to provide business-friendly labels and support cleaner filtering and analysis.

The raw incident data uses anonymized system identifiers such as Category 42, Subcategory 223, and Group 20. Business-friendly names were introduced through the dimension tables for portfolio presentation.

3.3 Initial Data Profiling

Before transforming the data, the incident table was reviewed in **Power Query** to understand its structure and identify potential data-quality issues.

The profiling process focused on:

- Row and column counts
- Error values

- Null and empty values
- Placeholder values
- Distinct and unique values
- Minimum and maximum values
- Data types
- Numerical distributions
- Date/time fields
- Identifier fields
- Categorical fields

This followed the project's core approach of **understanding the data before changing it**, rather than blindly replacing unusual or missing values. The project methodology explicitly emphasizes reviewing inconsistencies first and validating the data before modeling.

The raw dataset contains fields such as reassignment_count, reopen_count, made_sla, category, subcategory, priority, assignment_group, resolved_at, closed_at, and resolution-time fields, making profiling particularly important because these fields drive many of the final business measures.

3.4 Data Type Corrections

Data types were reviewed and corrected where necessary before the analytical model was developed.

Important fields included:

Field	Intended Data Type
number	Text
incident_state	Text
active	Boolean
reassignment_count	Whole Number
reopen_count	Whole Number
sys_mod_count	Whole Number
made_sla	Boolean
caller_id	Text
opened_by	Text
opened_at	Date/Time
sys_created_at	Date/Time
sys_updated_at	Date/Time
contact_type	Text
location	Text
category	Text

Field	Intended Data Type
subcategory	Text
impact	Text
urgency	Text
priority	Text
assignment_group	Text
resolved_at	Date/Time
closed_at	Date/Time
Resolution_Time_Hours	Decimal Number
Closure_Time_Hours	Decimal Number
Resolution_to_Closure_Hours	Decimal Number

Correct data types were particularly important because the analysis relies on **date filtering, numerical aggregation, Boolean SLA logic, identifiers, and resolution-time calculations.**

The raw data confirms that the incident table contains both timestamp fields and pre-calculated hour-based measures, making type consistency important for reliable calculations.

3.5 Data Quality Considerations

Several data-quality considerations were identified during preparation.

Placeholder values

The raw dataset contains ? values in some categorical fields.

For example, raw records can contain:

?

where a category-related or descriptive value would normally appear.

These values were treated as **unknown/unavailable information**, rather than inventing a business classification that was not supported by the source data.

This was particularly important for fields such as category, subcategory, and other descriptive attributes.

Missing values

Missing values were reviewed before deciding how they should be handled.

The principle was:

Do not blindly replace missing information. Understand what the missing value represents first.

This prevented unsupported assumptions from being introduced into the analytical model.

Identifier values

The original dataset uses anonymized identifiers for several descriptive fields, including:

- Category IDs
- Subcategory IDs
- Assignment Group IDs

Rather than treating these system identifiers as business names, dedicated dimension tables were created to provide more meaningful labels for the portfolio dashboard.

For example:

Category 42 → **Identity & Authentication**

and:

Group XX → **Service Desk Operations**

These labels are **simulated business-friendly names for portfolio presentation**, rather than claims about the original organization's actual department structure.

Resolution-time values

Resolution time was an important analytical field because the project compares both **mean and median resolution time** and analyzes resolution-time bands.

The raw data contains Resolution_Time_Hours, Closure_Time_Hours, and Resolution_to_Closure_Hours, allowing the analysis to distinguish the time required to resolve an incident from the additional time between resolution and closure.

Small-group percentages

During dashboard development, we identified an important analytical issue with assignment-group percentages.

Very small groups could produce extreme reassignment rates, including **100%**, even when only a very small number of incidents were involved.

Rather than treating those values as equally meaningful as large groups, the reassignment analysis applies a **minimum incident-volume threshold** when identifying groups requiring attention.

This reduces the risk of drawing conclusions from unstable small-sample percentages.

3.6 Preparation Outcome

After profiling and preparation, the dataset was ready for analytical modeling with:

- Appropriate data types
- Clean and usable date/time fields
- Valid numerical fields
- Investigated missing and placeholder values
- Structured category and subcategory attributes
- Business-friendly category and assignment-group dimensions
- Resolution-time fields suitable for analysis
- Reassignment and reopen fields suitable for operational metrics
- A consistent incident-level transactional structure

The prepared data was then used to build the Power BI data model and develop reusable DAX measures.

The overall preparation process followed the project's core principle:

Clean → Model → Analyze

This was important because the reliability of the final SLA, resolution-time, reassignment, and workload analysis depends on the quality and structure of the underlying data. The project guidance explicitly treats data preparation, modeling, DAX accuracy, and validation as connected stages rather than separate tasks.

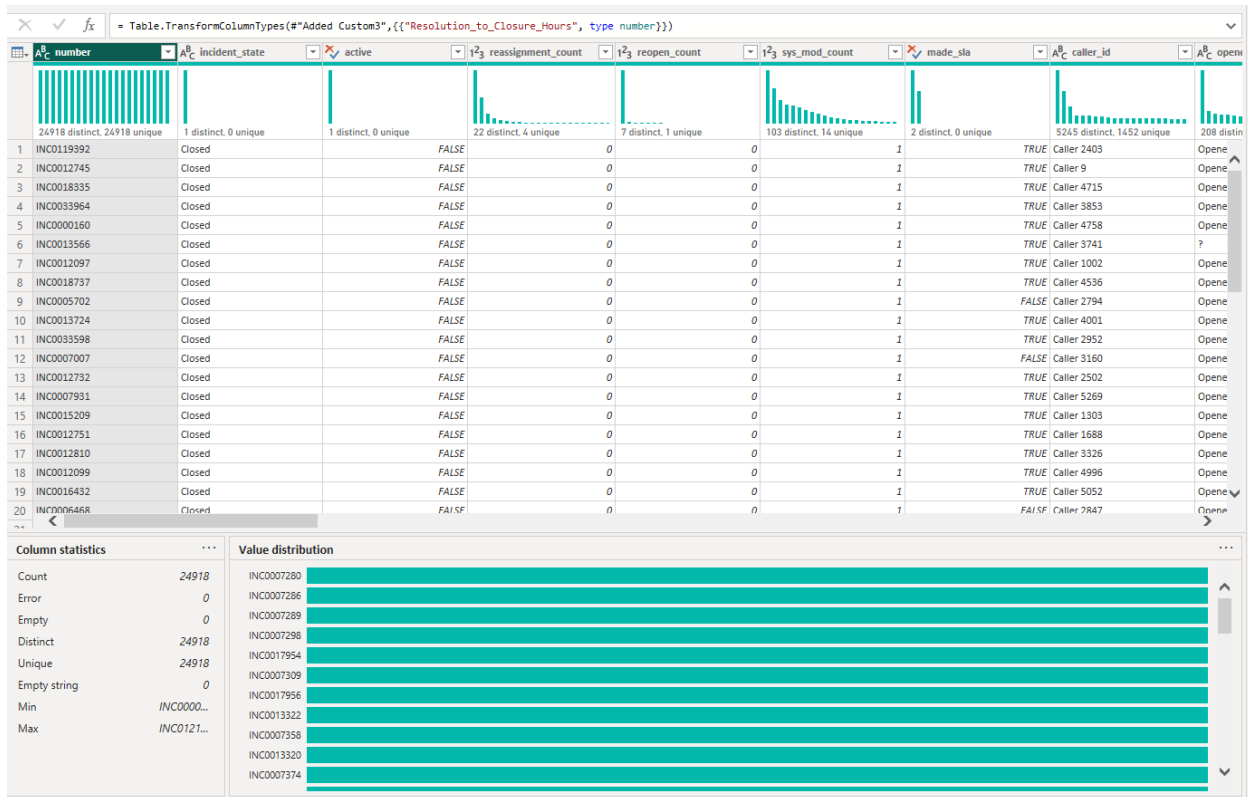


Figure 1— Initial Data Profiling in Power Query

4. Data Model

The Power BI solution uses a **multi-table analytical model** centered around the incident-level `Incidents` table.

Rather than keeping all information in a single flat table, the model separates the core incident records from descriptive dimensions, event/audit information, and specialized analytical tables.

This structure was designed to support the different analytical requirements of the project while keeping the model organized and easier to maintain.

4.1 Core Incident Table

Incidents

The `Incidents` table serves as the primary transactional table.

Each record represents an individual incident and contains the core information required for the analysis, including:

- Incident state and activity
- Priority
- Category and subcategory
- Assignment group
- SLA status
- Reassignment count
- Reopen count
- Opened, resolved, and closed timestamps
- Resolution and closure time

The table provides the foundation for the main measures used throughout the dashboard, including incident volume, SLA performance, resolution time, and reassignment rate.

4.2 Dimension Tables

Three dedicated dimension tables were created to provide descriptive context and more meaningful business-facing labels.

Dimension	Purpose
Dim_Category	Category identifiers and business-friendly names
Dim_Subcategory	Subcategory identifiers and business-friendly names
Dim_Assignment_Group	Assignment-group identifiers and business-friendly names

The original dataset uses anonymized identifiers such as:

```
Category 42
Subcategory 223
Group 70
```

Business-friendly names were added through these dimensions to make the dashboard more understandable to a non-technical audience.

These names are **simulated labels created for portfolio presentation** and do not represent verified names from the original organization.

4.3 Event, Audit & Analytical Tables

The model also contains supporting tables used for incident lifecycle analysis, validation, and specialized reporting.

Event & Audit Tables

- `incident_event_log`
- Audit - Final Incident Record

These provide additional incident activity and audit information that can support analysis and validation of the final incident records.

Analytical Tables

The model also includes purpose-built tables such as:

- `Category_Analysis`
- `Category_Resolution_Analysis`
- `Subcategory_Analysis`
- `Subcategory_Resolution_Analysis`
- `Contact_Channel_Analysis`
- `Priority_Resolution_Analysis`
- `SLA_Analysis`
- `SLA_By_Group`

These tables support specific analytical views including category workload, resolution performance, contact channels, priority performance, and assignment-group SLA analysis.

4.4 Model Design

The model was designed around the principle:

Keep each table focused on a specific analytical responsibility.

The `Incidents` table retains the detailed incident-level records, while dimensions provide descriptive context and supporting analytical tables handle specialized analysis.

Relationships were reviewed for:

- Cardinality
- Key uniqueness
- Filter propagation
- Relationship direction
- Duplicate values
- Appropriate table grain

This was particularly important because the final model contains substantially more tables and relationships than a simple reporting model.

4.5 Validation

The model and its outputs were validated throughout development rather than only after the dashboard was completed.

Key checks included:

- Overall incident count
- SLA totals
- Priority-level SLA results
- Reassignment totals
- Filter behavior
- Assignment-group calculations

The overall incident population was confirmed at:

24,918 incidents

Unexpected results were investigated against the underlying data before being accepted as business findings.

For example, the unusually low SLA results for High and Critical priorities were specifically checked against their underlying incident and SLA counts.

4.6 Model Outcome

The final model provides the foundation for three major analytical areas:

Service Performance

- SLA success
- Missed SLA
- Resolution time
- Priority performance

Workload

- Incident volume
- Category
- Subcategory
- Assignment group

Operational Behavior

- Reassignment
- Reopened incidents

- SLA vs. resolution performance
- Workload concentration

This structure allowed the dashboard to combine multiple perspectives while maintaining a consistent underlying incident dataset.

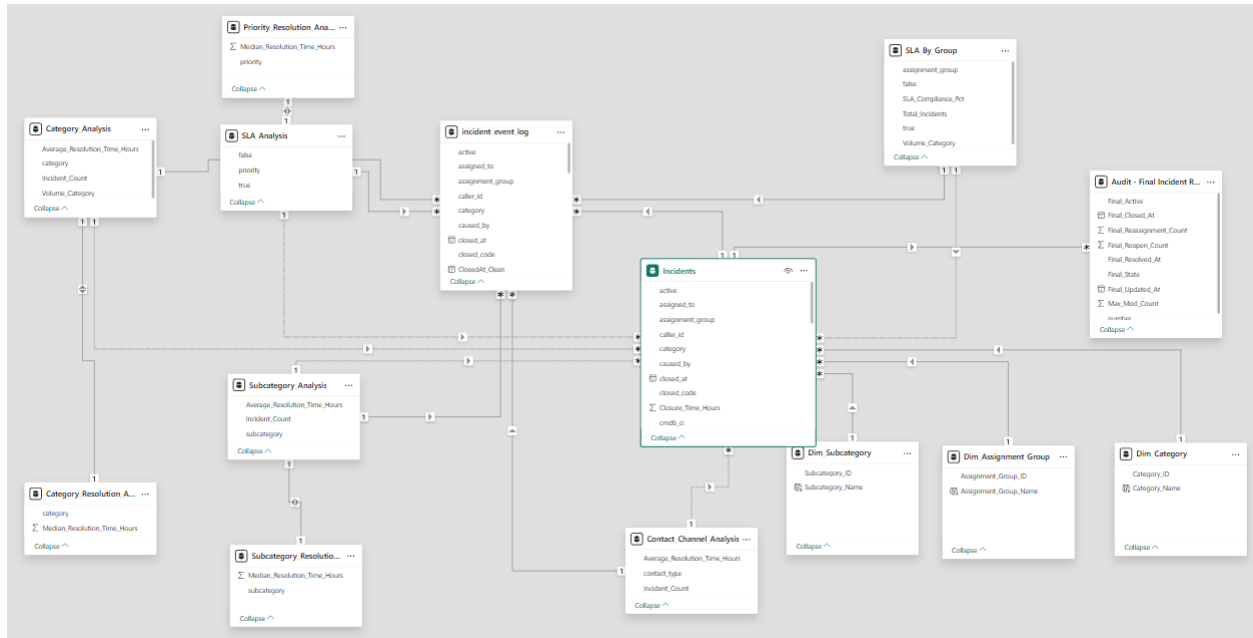


Figure 2— Power BI Data Model

5. Analytical Approach & DAX

After building the data model, the next stage was to create the analytical measures required to answer the business questions.

The analysis was built primarily using **reusable DAX measures**, allowing calculations to respond dynamically to the report's filter context.

The measures were organized around five main areas:

Analytical Area	Key Metrics
Workload	Incident Volume, Category, Subcategory, Priority
SLA Performance	SLA Success Rate, Missed SLA Rate
Resolution	Mean, Median, Resolution-Time Bands
Operational Activity	Reassignment Rate, Reopened Incidents

Analytical Area	Key Metrics
Group Performance	SLA, Resolution Time, Incident Volume

5.1 Incident Workload

Incident Volume

Incident Volume =
COUNTROWS(Incidents)

This measure counts incident records within the current filter context.

It serves as the foundation for the workload analysis and allows incident volume to respond dynamically to selections such as:

- Date Range
- Assignment Group
- Category
- Priority

The overall incident population is:

24,918 incidents

5.2 SLA Performance

SLA performance was one of the primary analytical areas of the project.

Two complementary measures were used:

SLA Success Rate

Measures the percentage of incidents that met the defined SLA.

63.42%

Missed SLA Rate

Measures the percentage of incidents that did not meet SLA.

36.58%

Together, these provide a complete view of SLA performance:

SLA Outcome	Result
Made SLA	63.42%
Missed SLA	36.58%

The dashboard also compares performance against a **70% SLA target**.

This allows management to see not only the current performance level but also the gap between actual performance and the target.

5.3 Resolution-Time Analysis

Resolution efficiency was evaluated using both **mean and median resolution time**.

Mean Time to Resolve

178.17 hours

Median Time to Resolve

22.10 hours

The substantial difference between the two indicates that the resolution-time distribution is highly skewed, with a smaller number of long-running incidents having a significant effect on the average.

For this reason, both measures were retained in the dashboard rather than relying on the mean alone.

Resolution time was also grouped into operational bands:

- Under 1 Hour
- 1–24 Hours
- 1–3 Days
- 3–7 Days
- Over 7 Days
- Unknown

This provides a clearer view of where incidents accumulate within the resolution lifecycle.

5.4 Reassignment Analysis

Reassignment was used as an indicator of operational activity and potential routing or ownership friction.

The core measure was:

```
Reassignment Rate =  
VAR TotalIncidents =  
    COUNTROWS (Incidents)  
  
VAR ReassignedIncidents =  
    CALCULATE (  
        COUNTROWS (Incidents),  
        FILTER (  
            Incidents,  
            COALESCE (Incidents[reassignment_count], 0) > 0  
        )  
    )  
  
RETURN  
    DIVIDE (  
        ReassignedIncidents,  
        TotalIncidents  
    )
```

An incident is considered reassigned when:

```
reassignment_count > 0
```

The resulting overall reassignment rate is:

45.63%

This does not prove that reassignment causes poor performance, but it provides a useful signal for investigating:

- Initial routing
- Ownership
- Skill alignment
- Escalation processes

5.5 Reopened Incidents

The analysis also tracks reopened incidents as an additional operational signal.

The dashboard reports:

275 reopened incidents

A reopened incident does not automatically indicate poor service. However, it identifies incidents requiring additional activity after being considered resolved and can therefore support further investigation into resolution quality and recurring issues.

5.6 Priority Analysis

Priority was analyzed against SLA performance to understand whether service-level outcomes varied across incident severity.

The validated results were:

Priority	Incident Volume	SLA Success
1 - Critical	270	1.85%
2 - High	408	0.49%
3 - Moderate	23,466	64.54%
4 - Low	774	84.11%

The unusually low results for High and Critical priorities were specifically investigated against the underlying data.

Rather than automatically treating the values as a calculation error, the incident and SLA counts were checked and the results were confirmed.

This followed the analytical process:

Unexpected result → Investigate → Validate → Interpret

5.7 Assignment-Group Analysis

Assignment groups were evaluated using several measures rather than a single KPI:

- Incident Volume
- SLA Success Rate
- Average Resolution Time
- Reassignment Rate

The **SLA vs. Resolution Matrix** combines these perspectives by comparing SLA performance with resolution time while using incident volume as additional context.

A **70% SLA target** provides the main performance reference.

This allows groups to be identified as areas of stronger performance or areas requiring further investigation.

5.8 Filter Context & Dynamic Measures

Filter context was an important consideration when developing the DAX layer.

The dashboard's Date Range and Assignment Group filters were tested to ensure that measures responded correctly.

This was particularly important for dynamic measures such as:

- Top Incident Category
- Top Assignment Group
- Reassignment Rate
- SLA Performance
- Incident Volume

The goal was to ensure that the dashboard displayed **context-sensitive results rather than static values**.

5.9 Validation Approach

DAX calculations were validated using:

- Reconciliation against the overall incident population
- Supporting tables
- Filter testing
- Visual cross-checking
- Investigation of unexpected results

For example, the reassignment calculation was checked against the underlying counts before the **45.63%** rate was accepted.

This validation process helped ensure that the measures were not only mathematically valid but also behaved correctly within the Power BI model.

5.10 Key Analytical Results

The final analytical layer produced the following headline metrics:

Metric	Result
Total Incidents	24,918

Metric	Result
SLA Success Rate	63.42%
Missed SLA Rate	36.58%
Mean Resolution Time	178.17 hrs
Median Resolution Time	22.10 hrs
Reassignment Rate	45.63%
Reopened Incidents	275

These measures form the analytical foundation of the three dashboard pages.

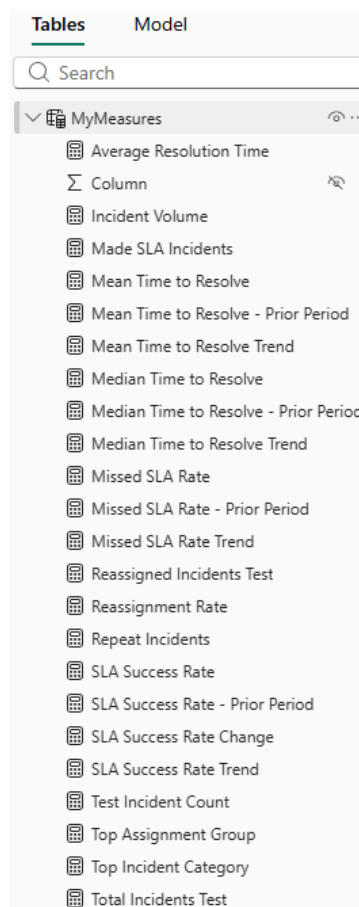


Figure 3— Core DAX Measures Used in the Analysis

6. Dashboard Analysis & Key Findings

The completed Power BI report was designed as a **three-page analytical experience**, with each page answering a different level of the business problem.

The dashboard progresses from **overall incident performance**, to **service-performance investigation**, and finally to **workload and operational analysis**.

6.1 Page 1 — Incident Command Center

Business Question

What is happening across the overall incident operation?

The Incident Command Center provides an executive-level overview of the incident environment.

It combines headline KPIs, incident trends, resolution-time distribution, SLA performance, priority distribution, and operational alerts into a single page.

Key Performance Indicators

Metric	Result
Total Incidents	24.918K
SLA Success Rate	63.42%
Mean Time to Resolve	178.17 hours
Repeat / Reopened Incidents	275

These metrics establish the overall scale of the workload and the current level of service performance.

Incident Volume Over Time

The incident-volume trend shows that incident activity is heavily concentrated in the earlier part of the analyzed period.

Incident volume declines substantially from May onward, with only a relatively small number of incidents occurring during the later months.

This pattern was verified against the underlying data rather than being treated as a visualization or calculation issue.

The analysis therefore retains the full available date range instead of artificially shortening the timeline.

Key takeaway

Incident demand was heavily concentrated in the earlier months of the dataset, while activity became substantially lower from May onward.

SLA Success Rate Over Time

The SLA trend provides a month-by-month view of SLA performance against the **70% SLA target**.

The results show that SLA performance fluctuates over time rather than remaining consistently above the target.

The overall SLA success rate is:

63.42%

This places overall performance below the 70% reference target.

Key takeaway

SLA performance does not consistently reach the 70% target, indicating an opportunity for further investigation into the drivers of SLA misses.

Resolution-Time Profile

The resolution-time profile compares incidents that **Made SLA** with those that **Missed SLA** across different resolution-time bands.

The bands include:

- Under 1 Hour
- 1–24 Hours
- 1–3 Days
- 3–7 Days
- Over 7 Days
- Unknown

The visual demonstrates that resolution duration is an important part of the SLA story, particularly among longer-running incidents.

This supports the use of both mean and median resolution time in the dashboard.

SLA Performance Distribution

Overall:

15.803K incidents — 63.42% Made SLA

9.115K incidents — 36.58% Missed SLA

This provides a simple executive-level view of the balance between successful and missed SLA outcomes.

Key takeaway

More incidents met SLA than missed it, but the 36.58% missed-SLA population remains significant enough to warrant operational attention.

Incident Priority Breakdown

The priority distribution is heavily concentrated in:

3 - Moderate

which accounts for approximately **94.17% of incidents**.

The remaining priorities represent much smaller portions of the overall incident population.

This distribution is important when interpreting the priority-level SLA results on Page 2 because the performance of a small priority population should not be interpreted in the same way as the dominant Moderate population.

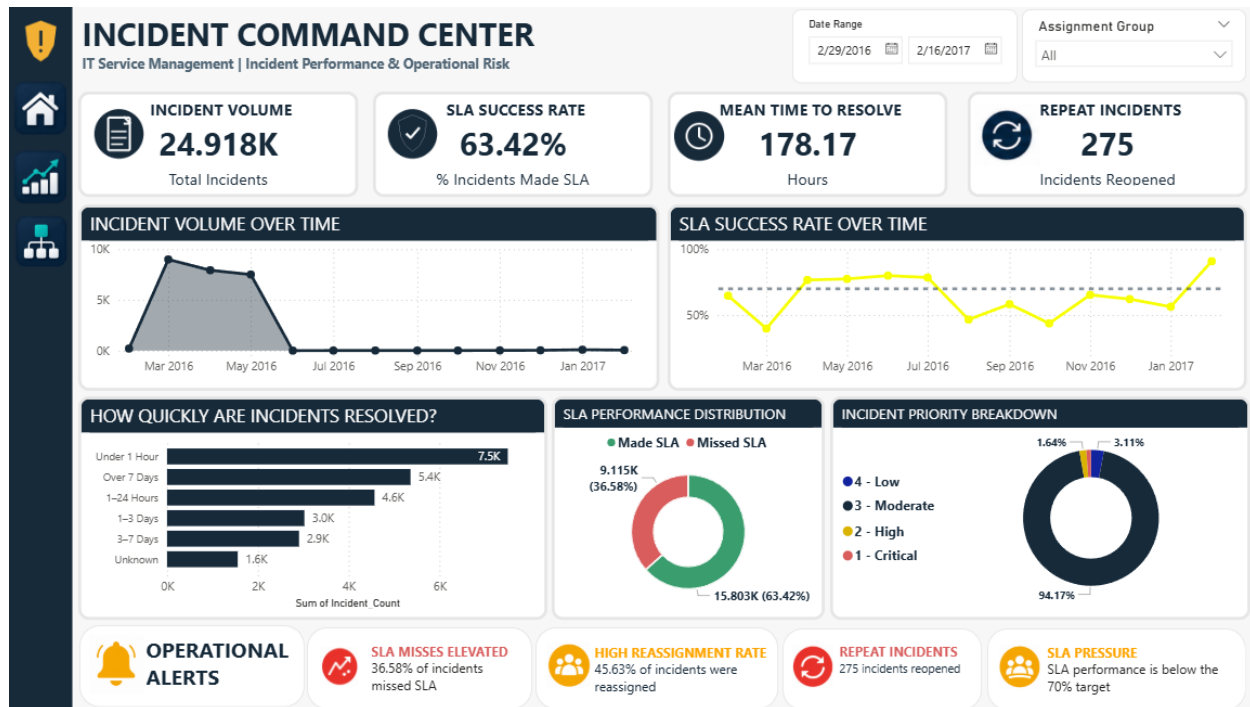


Figure 4— Incident Command Center

6.2 Page 2 — Service Performance Deep Dive

Business Question

Where is service performance under pressure, and which groups or priorities require closer investigation?

Page 2 moves from the executive overview into detailed service-performance analysis.

Key Metrics

Metric	Result
SLA Success Rate	63.42%
Missed SLA Rate	36.58%
Mean Time to Resolve	178.17 hours
Median Time to Resolve	22.10 hours

Mean vs. Median Resolution Time

One of the strongest findings in the report is the difference between:

Mean = 178.17 hours

and:

Median = 22.10 hours

The median is substantially lower than the mean, indicating that resolution times are strongly influenced by a smaller number of long-running incidents.

Key takeaway

The average resolution time alone does not represent the typical incident experience; long-running incidents materially increase the mean.

This is why both metrics are presented rather than relying on the mean alone.

SLA vs Resolution Matrix

The central visual on Page 2 compares assignment groups using three dimensions:

- **SLA Success Rate**
- **Average Resolution Time**
- **Incident Volume**

The dashboard also applies a:

70% SLA Target

reference line.

Assignment groups are interpreted using four operational zones:

☐ **Healthy**

High SLA success rate with faster resolution times.

☐ **Watch**

High SLA success rate but slower resolution times.

☐ Risk

Lower SLA success rate but relatively faster resolution.

● Critical

Lower SLA success rate combined with slower resolution.

This approach provides more useful operational context than ranking groups by SLA alone.

Key takeaway

Assignment-group performance should be evaluated using both SLA achievement and resolution speed, while incident volume provides additional context for prioritization.

SLA Performance by Priority

The validated priority results were:

Priority	Incident Volume	SLA Success Rate
1 - Critical	270	1.85%
2 - High	408	0.49%
3 - Moderate	23,466	64.54%
4 - Low	774	84.11%

The very low SLA rates for High and Critical incidents were specifically investigated and confirmed against the underlying data.

Key takeaway

SLA performance varies dramatically by priority, with High and Critical incidents showing substantially lower SLA success rates than Moderate and Low incidents in the analyzed data.

These results should be treated as an **investigation signal**, rather than proof of the underlying cause.

Areas Requiring Attention

The assignment-group table allows management to identify groups with weaker SLA performance while also considering:

- Average resolution time
- Incident volume

Conditional formatting makes lower-performing groups easier to identify.

The table therefore acts as a bridge between the high-level matrix and more detailed operational investigation.

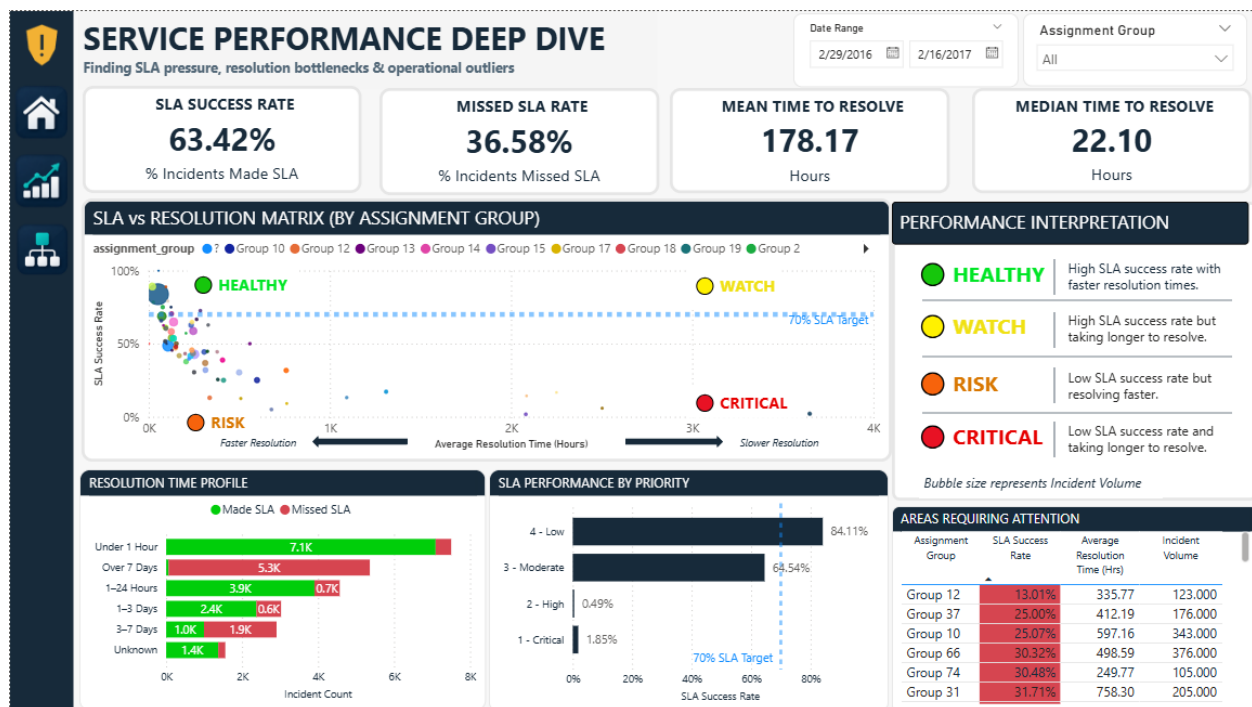


Figure 5— Service Performance Deep Dive

6.3 Page 3 — Incident Root Cause & Workload Analysis

Business Question

Where does incident demand originate, and where is operational effort concentrated?

Page 3 focuses on workload composition and operational behavior rather than repeating the SLA analysis from Page 2.

Incident Volume by Category

The largest incident categories include:

1. **Identity & Authentication** — approximately **3.6K**
2. **Software Support** — approximately **3.3K**
3. **Access & Permissions** — approximately **2.7K**
4. **Database Services** — approximately **2.4K**
5. **Network Services** — approximately **1.5K**

This indicates that incident demand is concentrated in a relatively small number of categories.

Key takeaway

Identity, software, access, and database-related services represent major sources of incident workload and therefore provide potential areas for targeted improvement.

Top 5 Incident Subcategories

The dashboard further breaks workload down into the highest-volume subcategories.

The leading subcategories include:

- **Password & Account Issues**
- **System Performance**
- **Login & Authentication**
- **Software Installation**
- **Email Issues**

The purpose of this analysis is to move beyond broad categories and identify the more specific sources of recurring incident demand.

Key takeaway

High-volume subcategories provide potential opportunities for targeted prevention, self-service, documentation, or process improvement.

Reassignment Rate by Assignment Group

The overall reassignment rate is:

45.63%

The assignment-group analysis identifies groups with high reassignment rates while applying a minimum incident-volume threshold to reduce the effect of extremely small groups.

Examples of groups appearing near the top of the analysis include:

- **Access Management** — 97.22%
- **Technical Operations** — 95.83%
- **IT Operations Support** — 91.67%
- **Platform Engineering** — 88.37%
- **Information Security** — 86.36%
- **Service Desk Level 2** — 85.71%

Key takeaway

Reassignment is a significant operational behavior in the dataset, with several sufficiently large assignment groups showing particularly high reassignment rates.

High reassignment should be treated as a signal for investigating routing, ownership, escalation, or skill-alignment processes rather than automatically being classified as a root cause.

Priority × Category Workload

The final matrix combines:

Category × Priority × Incident Volume

This allows management to see not only which categories generate the most incidents, but also how that workload is distributed across priority levels.

For example:

Identity & Authentication

- Critical — 34
- High — 33
- Moderate — 3,361
- Low — 130
- Total — **3,558**

Software Support

- Critical — 4
- High — 18

- Moderate — 3,244
- Low — 72
- Total — 3,338

The overall matrix reconciles to:

24,918 incidents

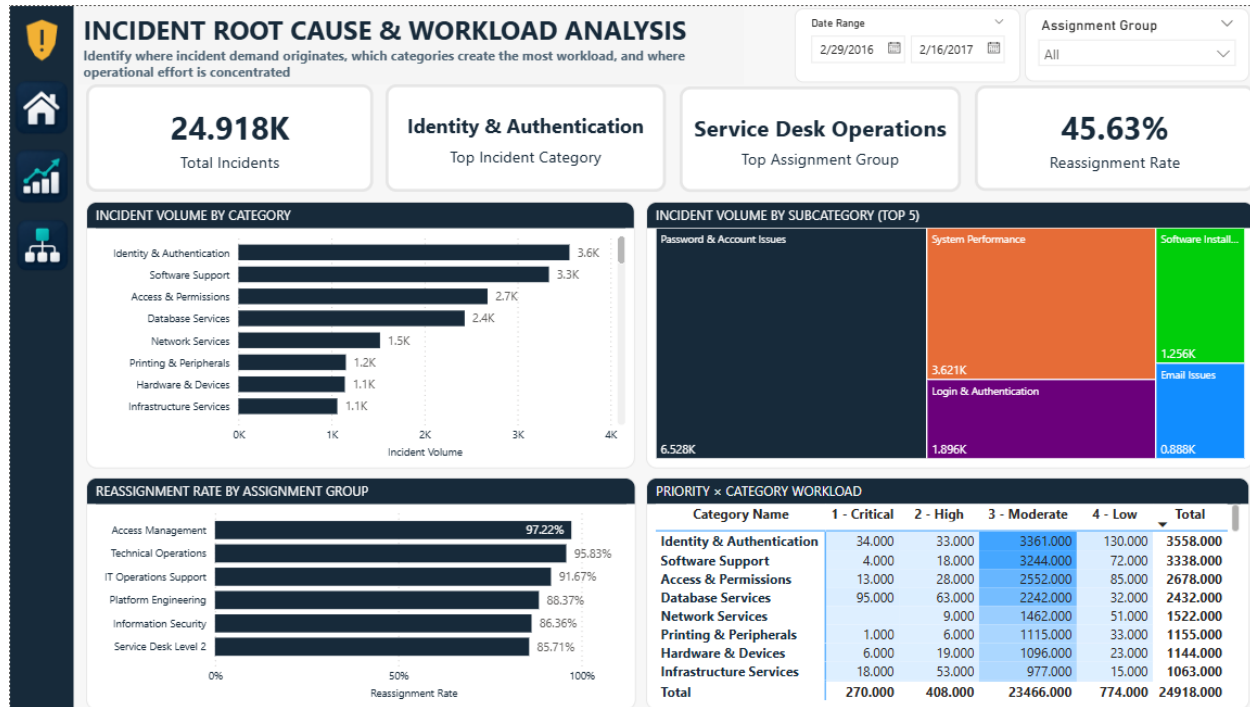


Figure 6— Incident Root Cause & Workload Analysis

6.4 Overall Analytical Conclusion

The dashboard demonstrates that incident performance should be evaluated as a **multi-dimensional operational problem** rather than through a single success metric.

The combination of:

Incident Volume + SLA Performance + Resolution Time + Priority + Category + Assignment Group + Reassignment

provides a more complete view of where service pressure is occurring and where management may want to investigate further.

The analysis does not attempt to claim causation where the data only demonstrates an association. Instead, the dashboard is designed to identify **patterns, outliers, and operational signals that can guide further investigation and improvement efforts.**

7. Recommendations & Business Actions

The analysis identified several areas where operational improvement could be explored.

1. Investigate SLA performance

Overall SLA success is **63.42%**, below the 70% target.

Management should investigate the assignment groups, priorities, and resolution-time bands associated with the highest SLA exposure.

2. Review High and Critical priority incidents

High and Critical incidents show substantially lower SLA success rates than Moderate and Low incidents.

These incidents should be reviewed to determine whether escalation processes, SLA definitions, workload, or resolution bottlenecks are contributing to the gap.

3. Investigate reassignment patterns

With **45.63% of incidents reassigned**, routing and ownership represent a significant area for investigation.

Groups with consistently high reassignment rates could be reviewed for:

- Initial routing accuracy
- Ownership clarity
- Skill alignment
- Escalation processes

4. Focus on high-volume categories

Categories such as **Identity & Authentication, Software Support, Access & Permissions, and Database Services** account for substantial incident workload.

Improvement initiatives in these areas could potentially affect a large portion of overall demand.

5. Address long-running incidents

The large difference between **mean resolution time (178.17 hours)** and **median resolution time (22.10 hours)** suggests that a smaller number of long-running incidents materially affect the average.

These incidents should be investigated separately to identify recurring bottlenecks.

Recommended Business Priorities

Priority	Recommended Action
High	Investigate SLA misses among High/Critical incidents
High	Identify causes of long-running incidents
Medium	Review high-reassignment assignment groups
Medium	Target high-volume incident categories
Ongoing	Monitor SLA performance against the 70% target

Overall Recommendation

The organization should prioritize **SLA improvement and long-running incident reduction**, while using reassignment and workload analysis to identify the operational areas most likely to benefit from deeper investigation.

The dashboard should be used as a decision-support tool to identify where investigation and improvement efforts should be focused, rather than as evidence of root cause by itself.

8. Limitations & Assumptions

This analysis was developed as a **portfolio case study using a public IT service management dataset**. Several limitations should therefore be considered when interpreting the results.

Dataset Limitations

- The dataset represents historical incident activity and may not reflect current IT service-management practices.

- Incident volume is heavily concentrated in the earlier part of the available period, with substantially fewer records later in the timeline.
- Some categories, priorities, and assignment groups have relatively small incident populations, which can produce extreme percentages.

Business Naming Assumptions

The original dataset contains anonymized identifiers such as Category 42 and Group 70.

Business-friendly names such as **Identity & Authentication** and **Service Desk Operations** were created for portfolio presentation. These names are simulated and should not be interpreted as verified organizational names.

Analytical Assumptions

- An incident with `reassignment_count > 0` is classified as reassigned.
- An incident marked as having met SLA is classified as a successful SLA outcome.
- Reopened incidents are treated as a signal of repeat operational activity rather than definitive evidence of poor resolution quality.
- High reassignment, long resolution times, and SLA misses are treated as **investigation signals**, not proven root causes.

9. Conclusion

This project demonstrates how Power BI can transform incident-level IT service management data into an interactive tool for **performance monitoring, workload analysis, and operational decision support**.

The analysis identified several important signals, including a **63.42% SLA success rate**, **178.17-hour mean resolution time**, **22.10-hour median resolution time**, and **45.63% reassignment rate**.

Rather than focusing on a single KPI, the dashboard combines **SLA performance, resolution efficiency, incident volume, priority, category, subcategory, and assignment-group behavior** to provide a broader view of service operations.

The three-page report provides a progression from:

Overall performance → Service performance → Workload & operational pressure

The project also demonstrates the importance of **data validation, filter context, appropriate dimensional modeling, and careful interpretation of unusual results**.

Ultimately, the dashboard is designed to help stakeholders answer a practical question:

Where should we focus our operational improvement efforts, and why?

The analysis provides the evidence needed to identify those areas and establishes a foundation for deeper investigation, process improvement, and future predictive analysis.